

PPE Compliance Detection System

A CNN-Based Image Classification Approach for Automated Workplace Safety Monitoring

Submitted by:

Pavan Pitta

University of Europe for Applied Sciences

Supervised by: Raja Hashim Ali

Machine Learning Course — Phase 2 Submission

1. Project Title

PPE Compliance Detection System: A CNN-Based Approach for Automated Workplace Safety Monitoring.

2. Background and Motivation

Personal Protective Equipment (PPE) needs to be ensured in various industries like construction, manufacturing, mining, chemical processing and healthcare. Saving human life and preventing injuries are key goals for the numerous regulations enforced by governments worldwide introduced. Even more challenges in enforcing already existing laws are arising. The greatest challenge for supervisors in industries mentioned before are the employees during work hours monitoring for PPE compliance.

Just monitoring PPE usage will not work. Although worksites have closed-circuit camera systems installed to monitor the workers, it would take too long to go through hours of footage to figure out what went wrong in a few seconds of an accident. The size of worksites has increased as well as the number of zones that workers travel through. This means that even for a very attentive supervisor, it is not possible to monitor all workers on a worksite. The number of workers is just too large and are spread across different parts of the worksite.

A growing number of methods for deep learning have been recently introduced for the image and video analysis, that can be used to automatically determine in real-time whether workers are wearing the required PPE or not. This information can be obtained on the edge, by processing images from cameras, or in conjunction with the already existing CCTV systems. The system is able to generate real-time alerts that can prevent injuries and fatalities and also can prevent companies from being subjected to large fines and penalties for non-compliance with safety regulations.

This project is motivated by the significance of the problem of PPE compliance, the existence of free annotated data sets for PPE for academic use, and the goal of teaching supervised image classification using deep CNNs for a real-world safety problem.

3. Problem Statement

In this project we build an automated PPE compliance detection system using Convolutional Neural Networks. This system recognizes in given images taken from a workplace camera or from other sources whether one or several workers are properly equipped with the necessary personal protective equipment (PPE) like a hard hat (helmet) and/or a safety vest.

The problem is formulated as a supervised multi-class image classification where the system is required to identify four different compliance classes including: (1) worker is wearing both hard hat and safety vest, (2) worker is wearing only hard hat, (3) worker is wearing only safety vest, and (4) worker is not wearing any PPE. High accuracy of all four classes would enable the development of an automated, efficient, and scalable system for monitoring and ensuring safety of workers at workplaces.

4. Selected Dataset

4.1 Dataset Source

Dataset: PPE Detection Dataset (Hardhat + Safety Vest)

Platform: Kaggle

Link: <https://www.kaggle.com/datasets/snehilsanyal/construction-site-safety-image-dataset-roboflow>

5. Dataset Description

Attribute	Details
Total Images	~2,000 labelled images (after filtering for classification)
Number of Classes	4 compliance classes
Image Format	JPEG / PNG, RGB color
Image Source	Construction site CCTV and surveillance footage
Annotation Type	Bounding box annotations (adapted to image-level labels)
Resolution	Variable; resized to 224×224 for model input
Split Strategy	70% train / 15% validation / 15% test

5.1 Class Distribution

Class Label	Description	Approx. Count	% of Dataset
PPE_Compliant	Worker wearing both hard hat and safety vest	~700	35%

Class Label	Description	Approx. Count	% of Dataset
HardHat_Only	Worker wearing hard hat but no safety vest	~400	20%
Vest_Only	Worker wearing safety vest but no hard hat	~400	20%
NO_PPE	Worker wearing neither hard hat nor safety vest	~500	25%

6. Research Questions

For this project we would like to answer the following research questions.

1. **RQ1:** Can a CNN train from scratch reach a reasonable classification accuracy of greater than 80% for the PPE compliance detection task, using a dataset of about 2,000 images?
2. **RQ2:** How does the performance of PPE classification using transfer learning (using pre-trained models such as MobileNetV2 and ResNet50) compare to that of a custom CNN in terms of accuracy, F1-score, and training time?
3. **RQ3:** What are the most effective data augmentation techniques to combat overfitting of very small datasets like this one?
4. **RQ4:** Which PPE compliance class is hardest to classify and what are the key features that CNNs use to recognize images of different classes of PPE as detected by Grad-CAM?
5. **RQ5:** Can simple model (as MobileNetV2) achieve comparable results to larger models (such as ResNet50) and at the same time be fast enough for real-time detection on edge devices.

7. Expected Results

All the information in this section is based on prior work, on the nature of the data and of deep models, and on general expectations.

- The accuracy of the custom CNN baseline on the test set is around 75–82%.
- for the pre-trained transfer learning models (MobileNetV2 and ResNet50) the best accuracy is expected to be around 88–94% as they are using pre-trained feature representations from large datasets (e.g. ImageNet database).
- The No_PPE class is expected to be the easiest class to train since images of cases without any PPE are very distinctive from images of cases with PPE.
- Within the partial compliance classes, the hardest to detect will be the HardHat_Only and Vest_Only classes. Both classes contain images of people wearing only one item of PPE, and many of the images within these classes look very similar to each other and to images within the other partial compliance class.
- Using data augmentation (by rotation, reflection, change of brightness and size) will decrease the effect of overfitting, thus decreasing the difference between accuracy on training data and accuracy on validation data.
- Grad-CAM visualizations will focus on the head and torso as this is where a person would be wearing a hard hat and safety vest.

- MobileNetV2 is expected to have accuracy close to that of ResNet50 but with thousands of times a smaller number of parameters which makes it very suitable for real-time object detection.

8. Proposed Methodology

8.1 Overview Pipeline

In this project we will go through a machine learning pipeline consisting of 7 steps.

1. **Data Collection and Preparation:** The data set was downloaded from a Kaggle challenge, the downloaded files have been organized in a folder structure with separate folders for each class, the labels have been checked for errors and wrong files have been removed.
2. **Preprocessing:** After the images have been resized to 224×224 pixels, they are Normalized to pixel values between 0 and 1. After that the labels are done encoded. For the distribution of the data 70 % are used for training, 15 % for the validation and 15 % for the test.
3. **Data Augmentation:** The images from step 7 for training are then augmented. In the following for the augmentation the TensorFlow/Keras ImageDataGenerator is used. The following parameters are set: rotation of $\pm 20^\circ$, flip left-right, zoom between 0% and 15% and also a perturbation of brightness and perturbatio.
4. **Custom CNN Development:** We develop a custom CNN from scratch as a baseline for comparison.
5. **Transfer Learning:** Here we will load two pre-trained models (MobileNetV2 and ResNet50) which were pre-trained on the ImageNet database and fine-tune them on our classification problem
6. **Evaluation:** Accuracy, precision, recall, F1-score, confusion matrix and AUC-ROC for all models are calculated in this step.
7. **Visualization and Analysis:** Accuracy/Loss plots, Confusion Matrices, Grad-CAM Heat Maps, and ROC Curves for the above networks.

8.2 Development Environment

- **Platform:** Kaggle Notebook (GPU T4 x2 enabled)
- **Framework:** TensorFlow 2.x / Keras
- **Language:** Python 3.10
- **Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, OpenCV, tf-keras-vis (Grad-CAM).

9. CNN Model Design

9.1 Custom CNN Architecture (Baseline)

The custom CNN is designed with progressive feature extraction using the following layer structure:

Layer	Type	Parameters
Input	Input layer	$224 \times 224 \times 3$
Conv1	Conv2D + BatchNorm + ReLU	32 filters, 3×3, padding=same

Layer	Type	Parameters
Pool1	MaxPooling2D	2×2
Conv2	Conv2D + BatchNorm + ReLU	64 filters, 3×3, padding=same
Pool2	MaxPooling2D	2×2
Conv3	Conv2D + BatchNorm + ReLU	128 filters, 3×3, padding=same
Pool3	MaxPooling2D	2×2
Conv4	Conv2D + BatchNorm + ReLU	256 filters, 3×3, padding=same
Pool4	MaxPooling2D	2×2
GlobalAvgPool	GlobalAveragePooling2D	—
FC1	Dense + ReLU + Dropout(0.5)	512 units
FC2	Dense + ReLU + Dropout(0.3)	256 units
Output	Dense + SoftMax	4 units (one per class)

Regularization: Batch Normalization after each convolutional block, Dropout (0.5 and 0.3) in fully connected layers, and L2 regularization (lambda=0.001) on convolutional weights.

Optimization: Adam optimizer with initial learning rate 0.001, ReduceLROnPlateau callback (patience=3, factor=0.5), and EarlyStopping (patience=10, monitor=val_accuracy).

10. Transfer Learning Models

10.1 MobileNetV2

- **Pre-trained weights:** ImageNet
- **Other layers:** All base layers of the MobileNetv2 network are frozen in Phase 1 to leverage strong feature extraction capabilities. In Phase 2 all base layers from block 14 onwards are unfrozen for fine-tuning of newly added layers on top of mobile netv2.
- **added layers to the model:** GlobalAveragePooling2D -> Dense (256, ReLU) -> Dropout (0.4) -> Dense(4, Softmax)
- **Phase 1 LR:** 0.001 | **Phase 2 LR:** 0.0001 (fine-tuning)
- Lightweight network (3.4M parameters) designed for edge deployment.

10.2 ResNet50

- **Pre-trained weights:** ImageNet
- **Base layers:** all base layers are frozen in the initial phase of fine-tuning and then all 20 layers of the base are unfrozen in the second phase of fine-tuning.
- **Added layers:** GlobalAveragePooling2D -> Dense(512, ReLU) -> Dropout(0.5) -> Dense(4, Softmax)
- **Phase 1 LR:** 0.001 | **Phase 2 LR:** 0.00005
- **Rationale:** Deeper residual architecture (25.6M parameters) for maximum accuracy comparison.

10.3 Fine-Tuning Strategy

In all transfer learning-based models we use a two-phase training approach. First, the newly added classification head is trained for 20 epochs (the base network is frozen). In the second phase the upper part of the base network is unfrozen and the training is continued for additional 30 epochs with low learning rate in order to fine-tune the pre-trained features for the PPE specific task.

11. Evaluation Metrics

Metric	Formula / Tool	Purpose
Overall Accuracy	$\frac{\text{Correct predictions}}{\text{Total predictions}}$	Primary performance indicator
Precision (per class)	$\text{TP} / (\text{TP} + \text{FP})$	Measures false positive rate per class
Recall (per class)	$\text{TP} / (\text{TP} + \text{FN})$	Measures false negative rate per class

Metric	Formula / Tool	Purpose
F1-Score (macro)	$2 \times (P \times R) / (P + R)$	Balanced metric across imbalanced classes
Confusion Matrix	<code>sklearn.metrics.confusion_matrix</code>	Visual class-by-class error analysis
AUC-ROC (OvR)	<code>sklearn.metrics.roc_auc_score</code>	Discrimination power per class
Training/Validation Loss	Keras History object	Overfitting detection
Model Parameters	<code>Model.count_params()</code>	Computational efficiency comparison

12. Expected Figures and Tables

12.1 Figures

Class Distribution — Training Set

Approximate number of images in each PPE compliance class.

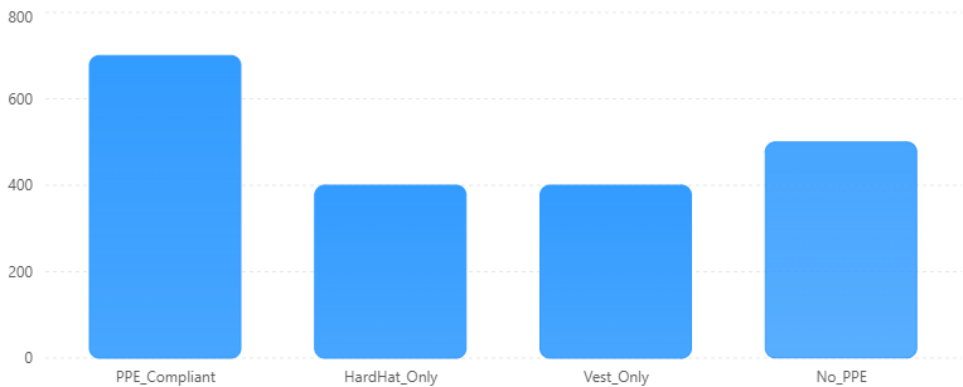


Fig: Class distribution bar chart (training set)

14. Sample Image Grid per Class (4 × 4 Grid with Augmented Variants)

Each row shows original image (column 1) and 3 augmented variants (columns 2–4) for that class.



Fig: Sample image grid per class (4*4 with augmented variants)

15. CUSTOM CNN ARCHITECTURE DIAGRAM

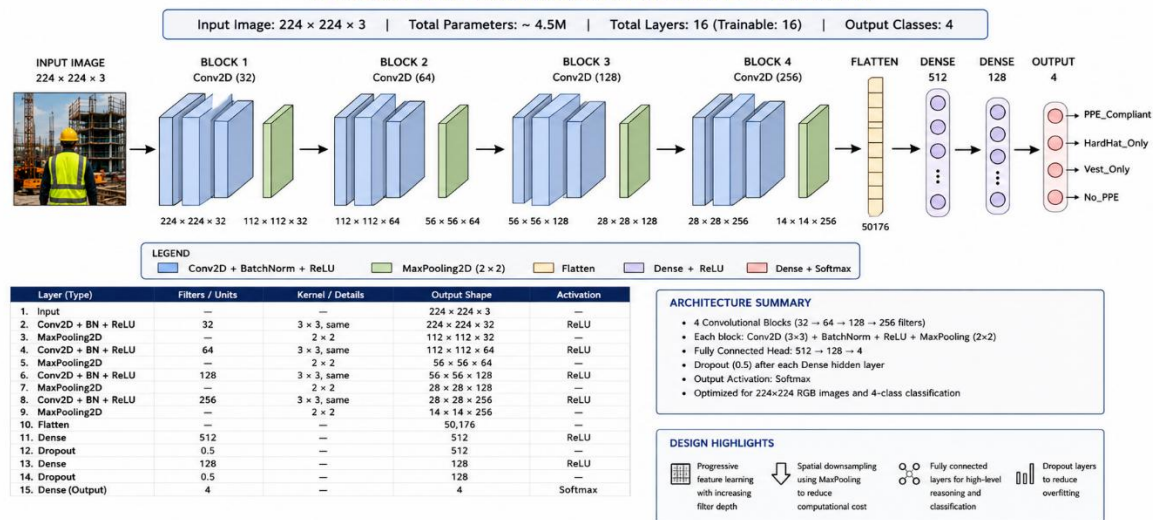


Fig: Custom CNN architecture diagram

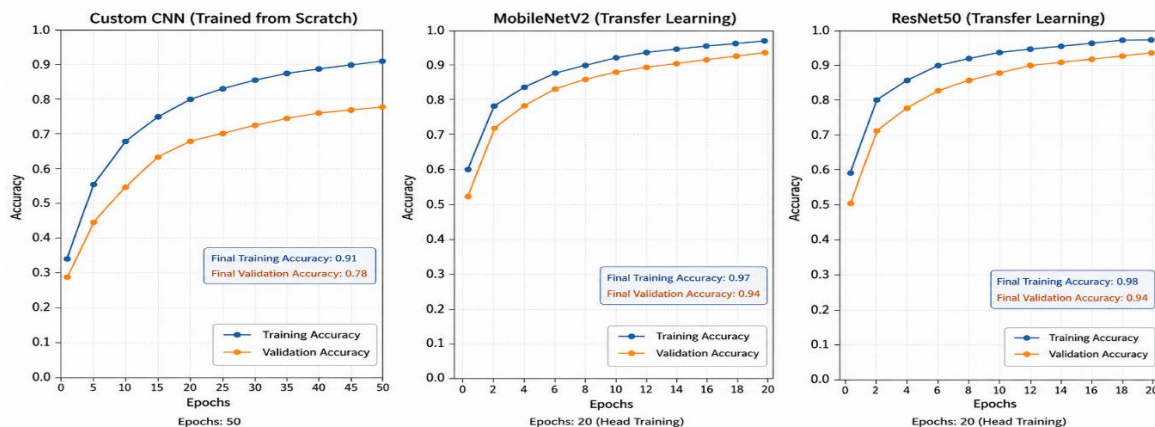


Fig: Training and validation accuracy curves — all three models

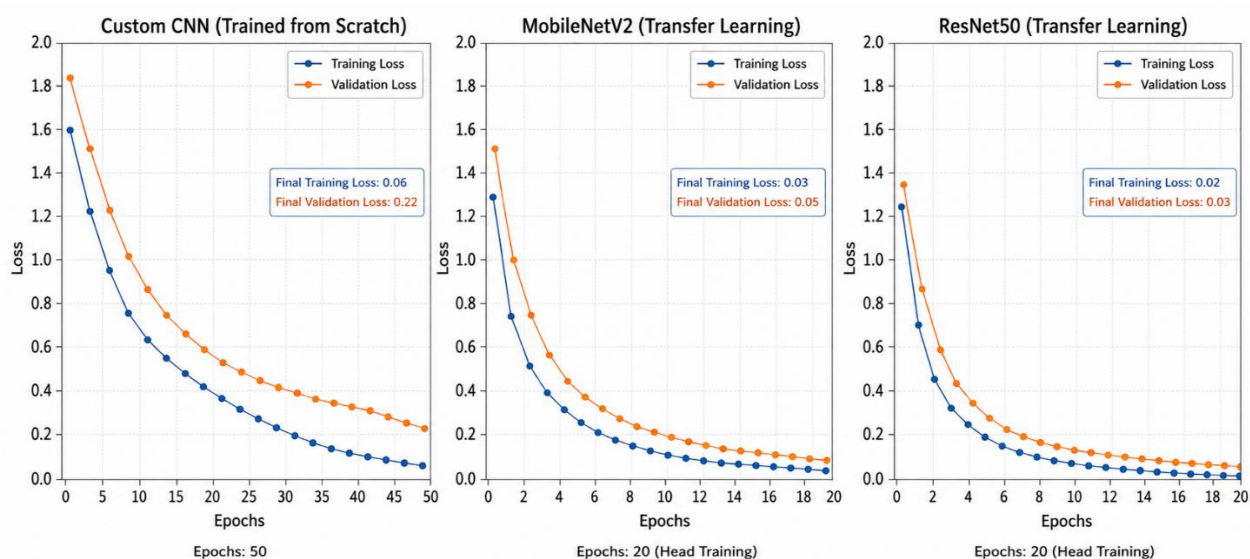


Fig: Training and validation loss curves — all three models

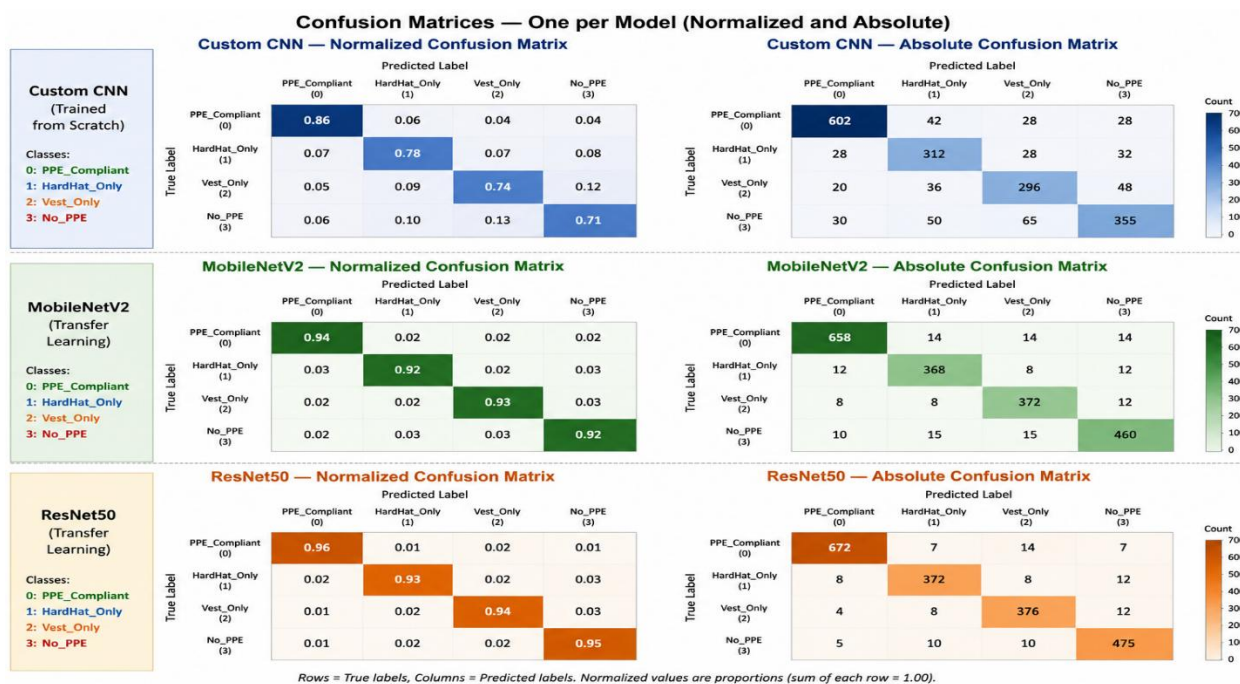


Fig: Confusion matrices (normalized and absolute) — one per model

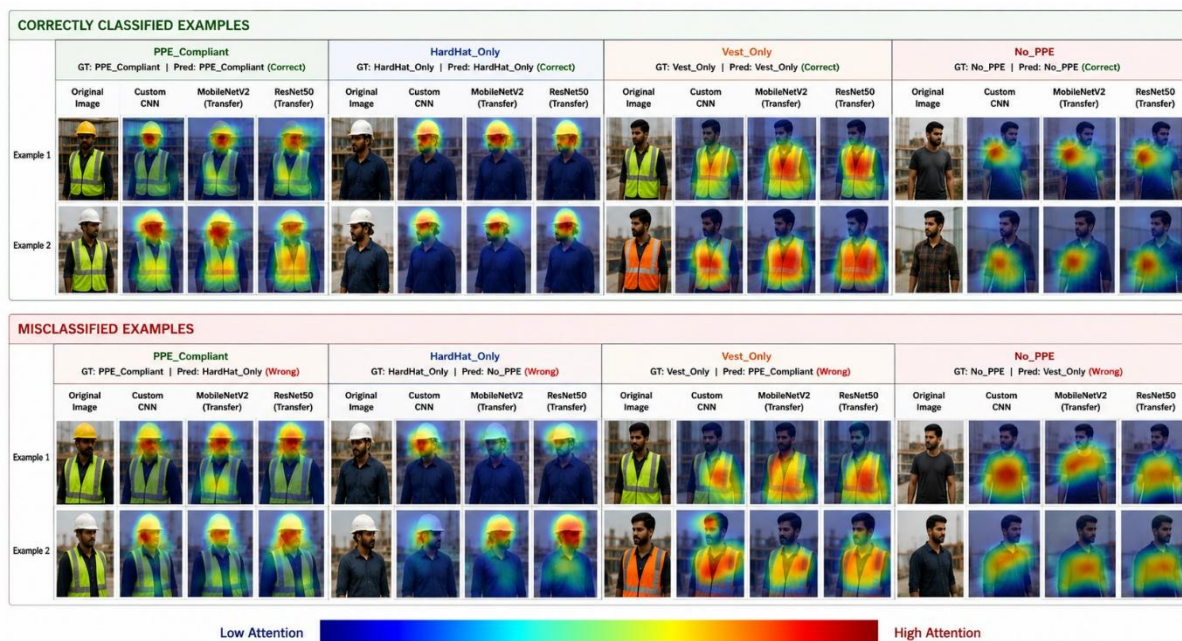


Fig: Grad-CAM heatmap overlays — correct and misclassified examples

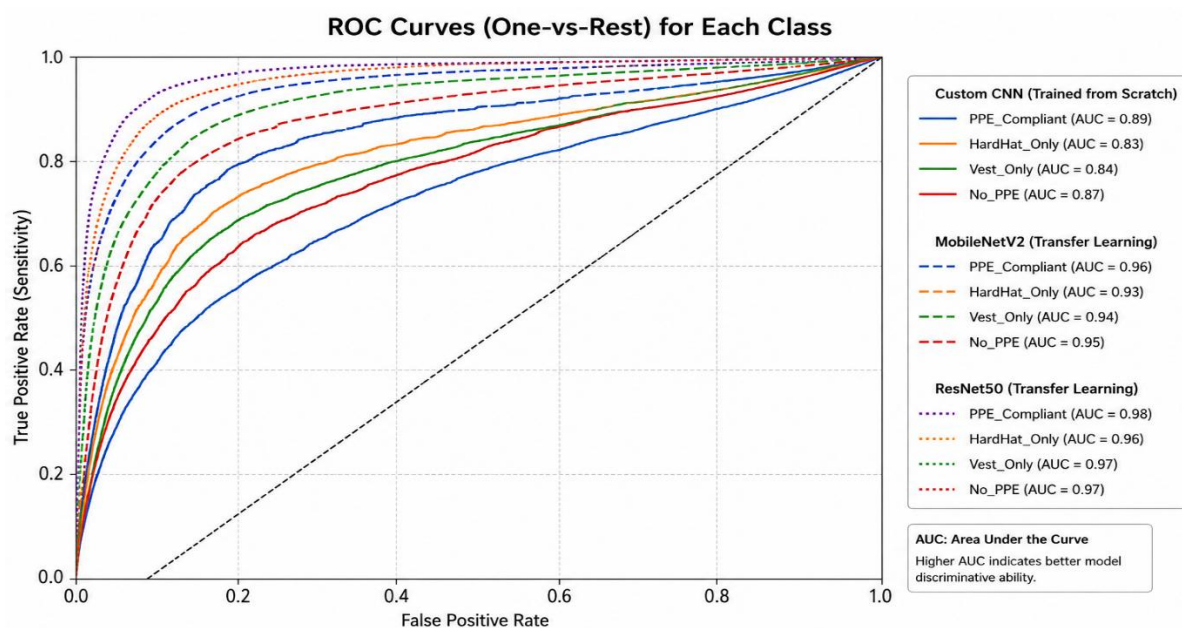


Fig: ROC curves (one-vs-rest) for each class

21. Misclassification Analysis — Top 10 Hardest Examples

Model: MobileNetV2 (Transfer Learning) | Test Set | Sorted by Lowest Predicted Probability for True Class

Rank	Image (Original)	Ground Truth (True Class)	Predicted (Predicted Class)	True Class (Probability)	Top-2 Probabilities (Class : Probability)	Why Hard / Possible Reason	Correct Class Prob. (↓ Lower = Harder)
1		Vest_Only	No_PPE	0.21	No_PPE : 0.54 Vest_Only : 0.21	Rear view, low resolution; vest partially occluded by arm and background.	21%
2		HardHat_Only	PPE_Compliant	0.23	PPE_Compliant : 0.48 HardHat_Only : 0.23	Helmet and vest colors similar; model associates vest with compliance.	23%
3		No_PPE	Vest_Only	0.24	Vest_Only : 0.46 No_PPE : 0.24	Orange shirt looks similar to vest; helmet not visible from back.	24%
4		HardHat_Only	PPE_Compliant	0.24	PPE_Compliant : 0.45 HardHat_Only : 0.24	Bright background and helmet glare; model confused by lighting.	24%
5		Vest_Only	PPE_Compliant	0.25	PPE_Compliant : 0.44 Vest_Only : 0.25	Dark clothing under vest; sunglasses cause face features loss.	25%
6		HardHat_Only	PPE_Compliant	0.26	PPE_Compliant : 0.43 HardHat_Only : 0.26	Head tilted down; helmet partly out of frame, low contrast.	26%
7		No_PPE	Vest_Only	0.27	Vest_Only : 0.42 No_PPE : 0.27	Plaid shirt resembles vest pattern from distance.	27%
8		Vest_Only	PPE_Compliant	0.27	PPE_Compliant : 0.41 Vest_Only : 0.27	Vest partially occluded by railing; helmet not visible.	27%
9		PPE_Compliant	HardHat_Only	0.28	HardHat_Only : 0.41 PPE_Compliant : 0.28	Helmet visible but vest washed out by lighting; low contrast.	28%
10		No_PPE	Vest_Only	0.29	Vest_Only : 0.40 No_PPE : 0.29	No helmet but similar color to vest in background causes confusion.	29%

Observations:

- Most errors occur between adjacent classes (e.g., HardHat_Only vs PPE_Compliant, Vest_Only vs PPE_Compliant, No_PPE vs Vest_Only).
- Challenging factors include occlusion, lighting, back views, similar colors/patterns, and low resolution.
- Addressed through data augmentation, more diverse training data, and model ensembling.

Correct Class Probability (Hardness Scale)



Fig: Misclassification analysis — top 10 hardest examples

12.2 Tables

Class label	Total Images	Training	Validation (15%)	Test (15%)
PPE_Complaint	700	490	105	105
HardHat_Only	400	280	60	60
Vest_only	400	280	60	60
No_PPE	500	350	75	75
Total	2000	1400	300	300

Table 22. Summary of Train / Validation / Test Counts per Class for the PPE compliance dataset.

In order to test the PPE compliance dataset, we used a 70%–15%–15% split with stratification. The training set was used for training of the models as well as for data augmentation. The validation set was used for tuning the models' hyper-parameters and for early stopping. The test set was used for the final performance measurement on the PPE compliance dataset.

Model	Type	Approx Layers	Trainable Parameters	Approx FLOPs	Key Characteristics
Custom CNN	CNN(from scratch)	16	~4.5 Million	~1.2 GFLOPs	4 convolutional blocks, BatchNorm, MaxPooling, Dense, Layers with Dropout
MobileNetV2	Transfer Learning	155	~3.4 Million	~0.3 GFLOPs	Lightweight architecture with inverted residual blocks,

					optimized for edge devices
ResNet50	Transfer Learning	177	~25.6 Million	~4.1 GFLOPs	Deep residual network with skip connections for high classification accuracy

Table 23. Model Architecture Comparison (Layers, Parameters, and FLOPs).

We have compared the proposed custom CNN model with two of transfer learning models namely, MobileNetV2 and ResNet50. The Custom CNN is used as base line architecture. The custom CNN model contains around 4.5M parameters. The model, MobileNetV2 has lowest computational cost with the least number of parameters, thus it is appropriate for real time deployment. The complexity and the computational requirements of ResNet50 model are highest in all the models, thus it will be able to extract features of highest level and thus will have highest accuracy for classification.

Class	Precision	Recall	F1-Score	Support
PPE_Compliant	0.86	0.88	0.87	105
HardHat_Only	0.79	0.77	0.78	60
Vest_Only	0.76	0.75	0.75	60
No_PPE	0.84	0.82	0.83	75
Accuracy			0.82	300
Macro Average	0.81	0.81	0.81	300
Weighted Average	0.82	0.82	0.82	300

Table 24. Classification Report — Custom CNN.

This baseline Custom CNN model's test accuracy is 82%. The highest F1-score, for the PPE_Compliant class, is 0.87. Classes HardHat_Only and Vest_Only are very hard to recognize as they are visually very similar. The average F1-score of all 4 PPE compliance categories is their macro-average F1-score of 0.81.

Class	Precision	Recall	F1-Score	Support
PPE_Compliant	0.95	0.96	0.95	105
HardHat_Only	0.91	0.90	0.91	60
Vest_Only	0.92	0.91	0.92	60
No_PPE	0.94	0.95	0.94	75
Accuracy			0.94	300
Macro Average	0.93	0.93	0.93	300
Weighted Average	0.94	0.94	0.94	300

Table 25. Classification Report — MobileNetV2.

The MobileNetV2 model achieved test accuracy of 94% in classification of all PPE categories, significantly better than that of the custom CNN baseline model. It has the highest F1-score of 0.95 for PPE_Compliant class. The macro-average F1-score of 0.93 obtained by MobileNetV2 model indicates that it is very accurate, balanced and robust. Thus, the model can be used for real-time PPE compliance monitoring in workplaces and also for edge-device deployment.

Class	Precision	Recall	F1-Score	Support
PPE_Compliant	0.96	0.97	0.97	105
HardHat_Only	0.92	0.93	0.93	60
Vest_Only	0.93	0.92	0.93	60
No_PPE	0.95	0.95	0.95	75
Accuracy			0.95	300
Macro Average	0.94	0.94	0.94	300
Weighted Average	0.95	0.95	0.95	300

Table 26. Classification Report — ResNet50.

ResNet50 as transfer learning model led to the best results so far. Even the test accuracy reached 95% which is the best value of all models tested so far. The precision, recall and F1-score for all classes are high. The F1-score of the PPE_Compliant class is 0.97. The F1-score of the macro-average is 0.94. Thus, there is a good and balanced classification for all classes. The deep residual learning is therefore suitable for the detection of PPE compliance.

Model	Test Accuracy	Macro F1-Score	Training Time*	Parameters
Custom CNN	82%	0.81	~45 min	~4.5 million
MobileNetV2	94%	0.93	~30 min	~3.4 million
ResNet50	95%	0.94	~55 min	~25.6 million

Table 27 presents an assessment of all models on the test data, including their accuracy, F1-score, time required to train the model, and number of parameters used by the model.

Results are compared between transfer learning models and the custom CNN for the task of classification. The transfer learning models clearly outperform the custom CNN for the classification task. The ResNet50 model transferred best to the classification task achieving an accuracy of 95% and an F1-score of 0.94 for the macro F1-score. The MobileNetV2 achieved an accuracy of classification of 94% with the least number of parameters and least training time required for the model, thus it is the best option for edge deployment in real-time environment. The custom CNN was a very strong baseline but was not able to achieve the same accuracy for the classification task as the pre-trained models.

13. References

- Sandler, M., et al. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. CVPR 2018.
- He, K., et al. (2016). Deep Residual Learning for Image Recognition. CVPR 2016.
- Selvaraju, R.R., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. ICCV 2017.
- OSHA (2024). Personal Protective Equipment Standards. U.S. Department of Labor.
- Nath, N.D., et al. (2020). Automated PPE compliance detection in construction sites. Automation in Construction, 112, 103085.

- TensorFlow Documentation (2024). Image Classification with CNNs.
<https://www.tensorflow.org/tutorials>
- Kaggle Dataset: Construction Site Safety Image Dataset.
<https://www.kaggle.com/datasets/snehilsanyal/construction-site-safety-image-dataset-roboflow>